

A Simple Motor-Parameter-Free Model Predictive Current Control for PMSM Drive

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Abstract—With the aim of improving parameter robustness in the model predictive current control (MPCC) for permanent magnet synchronous motor (PMSM) drive system, a simple motor-parameter-free model predictive current control (MPF-MPCC) methodology is proposed in this study. First, for different characteristics of dq-axes prediction model, the parameters of resistance and flux-linkage in the model are removed. Besides, an estimation factor is designed to substitute the inductance term in the model and is calculated based on a simple rhythm in real time. Utilizing the established predictive current model devoid of any motor parameters, the optimal voltage vector is accurately selected for controlling the motor. Finally, the effectiveness of the proposed MPF-MPCC methodology is substantiated through rigorous validation on a 1.25-kW experimental platform.

Index Terms—Model predictive current control (MPCC), motor-parameter-free, parameter robustness, permanent magnet synchronous motor (PMSM).

I. INTRODUCTION

PERMANENT magnet synchronous motor (PMSM) adopts power electronic frequency converter to realize frequency conversion speed regulation, which has the dominance of compact structure, heightened dependability, low maintenance cost and high efficiency [1], [2], [3]. At the present time, the control methodologies of PMSM drive system largely incorporate field-oriented control (FOC) and direct torque control (DTC) [4]. Despite the fact that FOC has been the dominant methodology to date, the linear proportional integral (PI) controllers used in some cases where no overshoot and high dynamics in required, are still inadequate [5], [6]. It is also important to note that DTC has several notable advantages, including a fast torque dynamic property and simple control structure, while the torque fluctuation of the motor is particularly pronounced at low speeds [7], [8], [9].

A. Literature Review

Recently, the ascension of model predictive control (MPC) within the realm of PMSM control can attributed to its inherent capabilities encompassing multitarget, multivariable, and multiconstraint control paradigms and intuitive design methodology [10], [11], [12]. Drawing upon distinct control behaviors, MPC diverges

into two branches that is finite control set MPC (FCS-MPC) and continuous control set MPC (CCS-MPC) [10], [13]. In FCS-MPC, the voltage vector is selected directly through the solution of the optimization problem without the complex modulation process [14]. Based on the classification of control objectives, MPC for PMSM drive can encompass model predictive current control (MPCC), model predictive voltage control (MPVC) and model predictive torque control (MPTC) [15], [16].

However, it is imperative to acknowledge that MPC is strongly reliant on the precision of the mathematical model of the controlled object [25]. Then, the notable consequence of the model parameter mismatch is the compromise in the accuracy of optimal voltage vector selection through cost function minimization, which in turn results in the perturbation of predicting future states [17]. Therefore, effectively improving the robustness of MPC is one of the present research hotspots and many excellent solutions have been proposed. In [18], the sliding mode observer (SMO) is combined with model reference adaptive control (MRAC) to address the jitter problem inherent in sliding mode control. H_∞ control is employed to guarantee the motor's steady-state controllability across a wide speed range and enhance its resilience to disturbance. In [19], a composite discrete-time SMO is employed within flux-linkage predictive control, grounded in the flux-linkage equation of state. This enhances the robustness of the system results from the flux parameter mismatch and rotor position error. Moreover, a lumped disturbance composite SMO is designed, which can quickly and accurately estimate the mismatch stemming from the motor parameters and compensate it to the prediction process to remove the influence generated by the motor parameter mismatch prediction process [20]. In [21], the error value of stator current prediction stemming from motor parameter mismatch is analyzed, and it is proposed that SMO and PI controller are combined to remove the stator inductance in real-time according to the prediction error, so as to remove the effect of inductance mismatch on the prediction state.

Moreover, numerous scholars have integrated the design of the observer with the enhancement of the conventional motor model, thereby enhancing its robustness. For instance, the literature enhances the conventional motor model by proposing an extended motor model that incorporates lumped disturbance. This is based on the conventional motor model. Then, the fast terminal SMO and fast terminal integral SMO are proposed to estimate the lumped disturbance that encompasses parameter mismatch disturbance and external disturbance [22]. Literatures [23] and [24] give the error form of the predicted current value under the condition of parameter mismatch, and then compensates the predicted current according to the error between the inaccurate predicted

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value and the measured value combined with the proportion factor to remove the impact of parameter mismatch. In [25], a MPCC methodology for PMSM grounded in a non-parametric prediction model is proposed. This methodology leverages current prediction differences, alongside sampling and storage of information to achieve precise current prediction.

Moreover, model free predictive control (MFPC) method has been extensively studied due to the advantage of not requiring any motor parameters [26], [27], [28], [29]. However, MPFC methods based on data-driven [27] are computationally intensive, which increases the hardware burden. Furthermore, MPFC methods based on ultra-local model often necessitate the utilization of observers to procure supplementary parameters within the model, such as extended state observe (ESO) [28] and SMO [29].

B. Motivation and Innovation

It should be noticed that the methodologies mentioned above have imperfections such as complex algorithms, need to design observers, high computational and need to identify multiple parameters, which undoubtedly makes the complexity of the control system be increased.

Aiming to enhance the parameter robustness of MPCC methodology in an effective manner without imposing an excessive time burden and reducing the number of system parameters that require tuning, a simple motor-parameter-free MPCC methodology is proposed in this study. The main contributions are as follows:

- 1) Based on different characteristics of dq-axes prediction model, parameters of resistance and flux-linkage are removed in the predictive model.
- 2) An estimation factor (M) that contains magnetic saturation, inverter nonlinearities, inductance, and all other factors reflecting current variations is designed, substituting the inductance term (T_s/L).
- 3) A simple control rhythm with low computing burden and with only one system parameter that udy is delineated as folloneed to be tuned is designed, enabling real-time accurately calculate the value of M .
- 4) A prediction model devoid of the existence in any motor parameters is established to achieve accurate current prediction.

C. Paper Organization

The organizational framework of this study is delineated as follows: Section II provides an exposition of the conventional MPCC methodology. Section III presents the proposed MPF-MPCC methodology. Section IV demonstrates the effectiveness of the proposed MPF-MPCC methodology. Finally, Section V concludes this study.

II. MATHEMATIC MODEL AND CONVENTIONAL METHODOLOGY

First, Section II is dedicated to detailing the mathematical modeling and the topological structure of PMSM drive system. Additionally, the conventional MPCC methodology is elaborated.

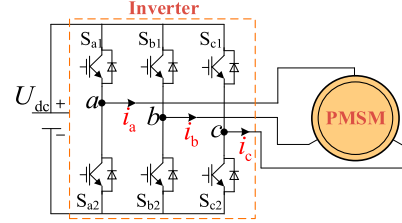


Fig. 1. Topology of PMSM system.

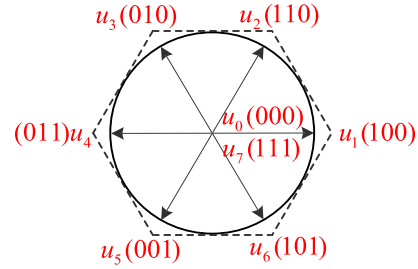


Fig. 2. Voltage vectors distribution diagram.

A. Mathematical Model and Topology of PMSM Drive System

The topology of PMSM drive system is illustrated in Fig. 1, where U_{dc} is represented as the dc-bus voltage and a two-level voltage source inverter is applied to this system. In addition, Fig. 2 depicts the eight basic voltage vectors generated by the two-level inverter, comprising six non-zero voltage vectors and two zero voltage vectors.

The voltage formula of PMSM for dq-axes is represented as follows:

$$\begin{cases} \frac{di_d}{dt} = \frac{1}{L_d} (u_d - Ri_d + \omega L_q i_q) \\ \frac{di_q}{dt} = \frac{1}{L_q} (u_q - Ri_q - \omega L_d i_d - \omega \psi_f) \end{cases} \quad (1)$$

where u_d and u_q represent voltages of dq-axes, while i_d and i_q represent their respective currents. The symbol ω is electrical angular velocity; ψ_f , L , and R represent flux-linkage, inductance, and resistance, respectively. For the surface-mounted PMSM researched in this study, the relationship between the inductance of the dq-axes is $L_d = L_q = L$.

Besides, the torque formula of PMSM for dq-axes is represented as follows:

$$T_e = \frac{3}{2} p i_q \psi_f \quad (2)$$

where p represents the number of pole pairs for the PMSM.

B. Conventional MPCC Methodology

Based on (1), the current prediction model of dq-axes is represented as follows:

$$\begin{cases} i_d^p(k+1) = i_d(k) - \frac{T_s}{L} G_d(k) + \omega T_s i_q(k) \\ i_q^p(k+1) = i_q(k) - \frac{T_s}{L} G_q(k) - \omega T_s i_d(k) \end{cases} \quad (3)$$

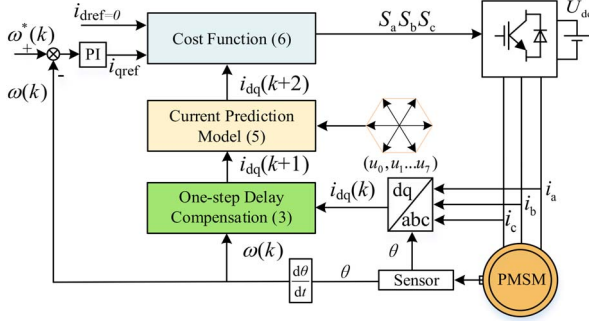


Fig. 3. Control schematic of conventional MPCC methodology.

where $\begin{cases} G_d(k) = R i_d(k) - u_d(k) \\ G_q(k) = R i_q(k) - u_q(k) + \omega \psi_f \end{cases}$ and T_s is the sampling period in the system.

Subsequently, within the conventional MPCC methodology, the optimal voltage vector is selected through the utilization of the cost function (4)

$$F = [i_{dref} - i_d^p(k+1)]^2 + [i_{qref} - i_q^p(k+1)]^2. \quad (4)$$

However, it is important to note that the presence of a one-step delay problem in the practical system will lead to inaccurate prediction results. Consequently, the one-step delay compensation must be incorporated into the prediction model through the addition of (3) and (4), which are then rewritten as (5) and (6)

$$\begin{cases} i_d^p(k+2) = i_d^p(k+1) - \frac{T_s}{L} G_d(k+1) + \omega T_s i_q^p(k+1) \\ i_q^p(k+2) = i_q^p(k+1) - \frac{T_s}{L} G_q(k+1) - \omega T_s i_d^p(k+1) \end{cases} \quad (5)$$

$$F = [i_{dref} - i_d^p(k+2)]^2 + [i_{qref} - i_q^p(k+2)]^2. \quad (6)$$

The control schematic of the conventional MPCC methodology is visually presented in Fig. 3, encompassing elements such as one-step delay compensation, current prediction model and cost function. The elements depicted in Fig. 3 reveal that the optimal voltage for the motor is determined by selecting the voltage vector that minimizes the cost function (6).

C. Analysis of Parameter Mismatch in Conventional MPCC Methodology

It is well known that MPC is a methodology that is sensitive to the motor parameter. As previously stated, the mismatched parameter (L , ψ_f , and R) will affect the predicted currents, thereby causing the incorrect voltage vector to be selected which in turn will result in a reduction in the controllability of the PMSM system [1]. This part presents a brief analysis of the error in current prediction of dq-axes within the conventional MPC methodology is briefly respectively analyzed when the parameters are mismatched.

In this study, L_0 , ψ_{f0} , and R_0 are the parameter of inductance, flux-linkage and resistance in the model. L , ψ_f , and R are the accurate parameter in ideal conditions. Besides, $\Delta\psi_f$, ΔL , and ΔR are the parameter errors, and the relationship among them is

$$L_0 = L + \Delta L, \quad \psi_{f0} = \psi_f + \Delta\psi_f, \quad R_0 = R + \Delta R. \quad (7)$$

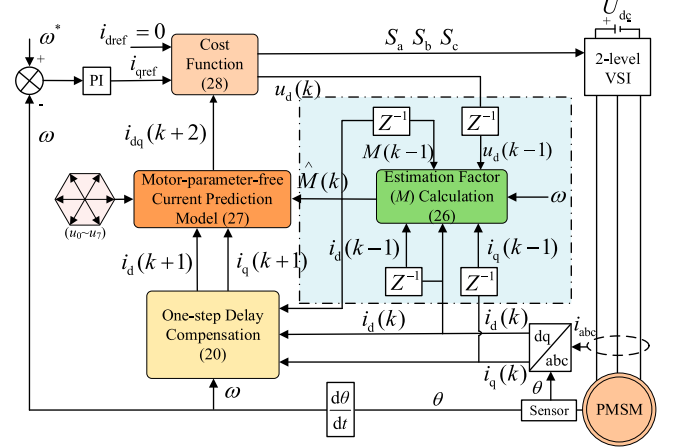


Fig. 4. Control diagram of the proposed MPF-MPCC methodology.

When the above of parameters are mismatched, the incorrect current prediction results can be acquired by (8)

$$\begin{cases} i_d^{\text{error}}(k+1) = i_d(k) - \frac{T_s}{L_0} G_d^e(k) + \omega T_s i_q(k) \\ i_q^{\text{error}}(k+1) = i_q(k) - \frac{T_s}{L_0} G_q^e(k) - \omega T_s i_d(k) \end{cases} \quad (8)$$

where $\begin{cases} G_d^e(k) = R_0 i_d(k) - u_d(k) \\ G_q^e(k) = R_0 i_q(k) - u_q(k) + \omega \psi_{f0} \end{cases}$.

As illustrated in (8), discrepancies in the parameters can result in alterations to the predicted outcomes, which may consequently affect the system control performance. This is evidenced by the observation that inductance and flux-linkage mismatch exert a significant influence on the controlling ability of PMSM systems, while the influence related to resistance mismatch is minimal [16], [21].

III. PROPOSED MPF-MPCC METHODOLOGY

With the objective of mitigating the adverse influences stemming from parameter mismatch in the dq-axes, an MPF-MPCC methodology is proposed in which the predicted currents of the next control period are obtained without any motor parameter. The control diagram of MPF-MPCC methodology is depicted in Fig. 4, encompassing four key elements: motor-parameter-free current prediction model, estimation factor (M) calculation, one-step delay compensation, and updated cost function.

D. Motor-Parameter-Free Current Prediction Model in d-Axis

By subtracting (8) from (3), the predictive errors of the d-axis current (E_d) are acquired:

$$\begin{aligned} E_d &= i_d^{\text{error}}(k+1) - i_d^p(k+1) \\ &= \frac{TR_0 \Delta L - TL_0 \Delta R}{L_0(L_0 - \Delta L)} i_d(k) - \frac{T \Delta L}{L_0(L_0 - \Delta L)} u_d(k). \end{aligned} \quad (9)$$

It can be seen from (9) that there has been an impact of inductance and resistance in the predictive error of the d-axis current.

When dealing with the resistance mismatch, the parameter errors stemming from inductance (ΔL) can be not considered. Thus, the predictive errors of d-axis current (9) will be updated to

$$E_d = -(T_s/L_0) \cdot \Delta R i_q(k). \quad (10)$$

The analysis in (10) indicates that the magnitude of T_s/L_0 is too small. Consequently, the predictive errors of d-axis current (E_d) can be approximately regarded as equal to zero ($E_d \approx 0$). Thus, the predictive errors of the d-axis current resulting from resistance mismatch can be ignored.

Therefore, the predicted current of the d-axis without considering the resistance at the $k+1$ control period is shown as

$$i_d^p(k+1) = A_d + (T_s/L) \cdot u_d(k) \quad (11)$$

where $A_d = i_d(k) + \omega T_s i_q(k)$ that contains the sampled information needed in d-axis.

E. Motor-Parameter-Free Current Prediction Model in q-Axis

Similar as the calculation process of (9), the predictive errors of q-axis current are acquired

$$\begin{aligned} E_q &= i_q^{\text{error}}(k+1) - i_q^p(k+1) \\ &= \frac{T_s R_0 \Delta L - T_s L_0 \Delta R}{L_0(L_0 - \Delta L)} i_q(k) - \frac{T_s \Delta L}{L_0(L_0 - \Delta L)} u_q(k) \\ &\quad + \frac{T_s \omega(\psi_{f0} \Delta L - L_0 \Delta \psi_f)}{L_0(L_0 - \Delta L)}. \end{aligned} \quad (12)$$

It can be seen from (12) that there has been influenced by inductance, resistance and flux-linkage in the predictive errors of q-axis current.

In instances where a resistance mismatch occurs, the errors stemming from the parameters of inductance and flux-linkage (ΔL and $\Delta \psi_f$) can be not considered. Thus, the predictive errors of q-axis current (12) can be revised to

$$E_q = -(T_s/L_0) \cdot \Delta R i_q(k). \quad (13)$$

Similarly, due to the relatively small magnitude of T/L_0 in (13), the predictive errors of q-axis current (E_q) can be approximately regarded as equal to zero ($E_q \approx 0$). Consequently, the errors in current prediction for q-axis resulting from resistance mismatch can be left out of consideration.

Thus, the predictive current of q-axis without considering resistance in the $k+1$ control period is obtained as

$$i_q^p(k+1) = i_q(k) + (T_s/L) \cdot u_q(k) - T_s \omega i_d(k) - (T_s/L) \cdot \omega \psi_f. \quad (14)$$

To remove the impact of the flux-linkage in (14), an incremental prediction model is applied in this study, which predicts current based on the difference of two neighboring instants [21].

Based on (14), the current of q-axis without considering resistance in the k control period is obtained as

$$i_q^p(k) = i_q(k-1) + (T_s/L) \cdot u_q(k-1) - T_s \omega i_d(k-1) - (T_s/L) \cdot \omega \psi_f. \quad (15)$$

Then, by subtracting (14) and (15), the model of the q-axis is represented as follows:

$$\Delta i_q^p(k+1) = i_q^p(k+1) - i_q^p(k). \quad (16)$$

By simultaneous (15) and (16) is updated as follows:

$$\Delta i_q^p(k+1) = \Delta i_q(k) - \omega T_s \Delta i_d(k) + (T_s/L) \cdot \Delta u_q(k) \quad (17)$$

$$\text{where } \begin{cases} \Delta i_{dq}(k) = i_{dq}(k) - i_{dq}(k-1) \\ \Delta u_q(k) = u_q(k) - u_q(k-1) \end{cases}$$

Then, by (17), the updated current prediction model of q-axis is obtained

$$i_q^p(k+1) = A_q(k) + (T_s/L) \cdot \Delta u_q(k) \quad (18)$$

where $A_q = 2i_q(k) - i_q(k-1) - \omega T_s \Delta i_d(k)$ that contains the sampled information needed in q-axis.

From the above analysis and coupling (11) and (18), the current prediction model of dq-axes is obtained

$$\begin{cases} i_d^p(k+1) = A_d(k) + (T_s/L) \cdot u_d(k) \\ i_q^p(k+1) = A_q(k) + (T_s/L) \cdot \Delta u_q(k). \end{cases} \quad (19)$$

As illustrated in (19), there is no parameter associated with the resistance and flux-linkage. This indicates that the mismatched between the two parameter has no effect on the current prediction model and the controllability of the drive system.

However, the inductance parameter mismatch of the motor still exists in (19) and affects the controllability of the drive system. To attain the current prediction devoid of the existence in any motor parameters, the estimation factor (M) is involved to substitute the inductance term (T_s/L). It considers magnetic saturation, inverter nonlinearities, inductance, and all other factors reflecting current variations.

Thus, the motor-parameter-free current prediction model is obtained

$$\begin{cases} i_d^p(k+1) = A_d(k) + M \cdot u_d(k) \\ i_q^p(k+1) = A_q(k) + M \cdot \Delta u_q(k) \end{cases} \quad (20)$$

where $M = T_s/L$.

In conclusion, the finding of (20) indicates that no motor parameters are present in the prediction model. Moreover, the current prediction model is so elementary that only the estimation factor (M) needs to be obtained. Moreover, the methodology of the calculation will be presented in the subsequent part.

F. Calculation of the Estimation Factor (M)

To facilitate the computation of the M , a simple control rhythm is devised and can be expressed as follows:

$$J = \underbrace{[i_d^p(k+1) - i_d(k+1)]^2}_{\text{Term 1}} + \lambda \underbrace{[\hat{M}(k) - M(k-1)]^2}_{\text{Term 2}} \quad (21)$$

where $i_d^p(k+1)$ and $i_d(k+1)$ represent the predicted current and the applied current of the d-axis and at the $k+1$ control period, respectively. Moreover, λ is a weight coefficient that balancing the current difference and the estimation factor difference.

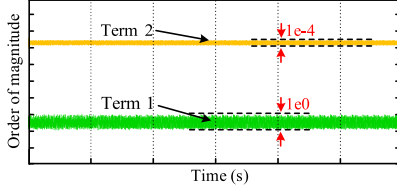


Fig. 5. Schematic diagram of the order of magnitude.

Fig. 5 shows the simulation results about the order of magnitude of terms 1 and 2. Observations from Fig. 5 that the order of term 1 is much smaller than that of term 2. Based on actual debugging in the project and the value of λ is set to $1e^3$ in this experiment.

By simultaneous (11) and (21), the (21) can be updated as follows:

$$J = [A_d(k) + M(k)u_d(k) - i_d(k+1)]^2 + \lambda[\hat{M}(k) - M(k-1)]^2. \quad (22)$$

The design goal of the control rhythm is to find $\hat{M}(k)$ that minimizes J . According to the necessary condition for taking the extreme value is that the partial derivative is zero, the partial derivative of $\hat{M}(k)$ concerning the control rhythm J is obtained as follows:

$$\frac{\partial J}{\partial \hat{M}(k)} = 2[A_d(k) + M(k)u_d(k) - i_d(k+1)] \cdot u_d(k) + 2\lambda[\hat{M}(k) - M(k-1)]. \quad (23)$$

Moreover, by making it equal to zero ($\partial J / \partial \hat{M}(k) = 0$), the result of the estimation factor $M(k)$ is obtained as follows:

$$\hat{M}(k) = M(k-1) - \frac{[A_d(k) + M(k)u_d(k) - i_d(k+1)]}{\lambda} \cdot u_d(k). \quad (24)$$

Equation (24) is shifted backward by one control period, subsequently yielding the estimation factor $M(k-1)$ at the $k-1$ instant as demonstrated:

$$\hat{M}(k-1) = M(k-2) - \frac{[A_d(k-1) + M(k-1)u_d(k-1) - i_d(k)]}{\lambda} \cdot u_d(k-1). \quad (25)$$

Because the mechanical time constant of PMSM is much larger than its electrical time constant, it can be assumed that the computed values of M are equal in neighboring control period when the computational process stabilizes, i.e.

$$\hat{M}(k) = M(k-1) - \frac{[A_d(k-1) + M(k-1)u_d(k-1) - i_d(k)]}{\lambda} \cdot u_d(k-1). \quad (26)$$

Through the above analysis, there are no motor parameters in the updated prediction model and in the process of calculate the estimated factor (M).

G. One-Step Delay Compensation

As the practical system encounter one-step delay, the updated current prediction model that accounts for one-step delay

compensation can be expressed as follows:

$$\begin{cases} i_d^p(k+2) = i_d^p(k+1) + T_s \omega i_q(k+1) + M u_d(k+1) \\ i_q^p(k+2) = 2i_q^p(k+1) - i_q(k) - \omega T_s \Delta i_d(k+1) + M \Delta u_q(k+1) \end{cases} \quad (27)$$

$$\text{where } \begin{cases} \Delta i_d(k+1) = i_d^p(k+1) - i_d(k) \\ \Delta u_q(k+1) = u_q(k+1) - u_q(k) \end{cases}.$$

The optimal voltage vector is determined by the cost function (28) that is updated from (6)

$$F^* = [i_{dref} - i_d^p(k+2)]^2 + [i_{qref} - i_q^p(k+2)]^2. \quad (28)$$

In the summary, the theoretical derivation of the proposed MPF-MPCC methodology is concluded.

H. Stability Analysis

According to the voltage equation of PMSM, the true response of PMSM can be expressed as

$$i_d^c(k+1) = \left(1 - \frac{T_s \cdot R}{L}\right) i_d(k) + \frac{T_s}{L} \cdot u_d(k) + T_s \cdot \omega_e i_q(k). \quad (29)$$

Combining (11) and (29), the relationship of $i_d^p(k+1)$ and $i_d^c(k+1)$ is as follows:

$$i_d^c(k+1) = \left(1 - \frac{T_s R}{L}\right) i_d(k) + \frac{T_s}{L} \cdot \frac{i_d^p(k+1) - A_d(k)}{M} + T_s \omega i_q(k). \quad (30)$$

Since M can be computed in real time by (26), bringing (26) to (30) and organizing it yields

$$\begin{aligned} i_d^c(k+1) &= \frac{\lambda T_s}{AL} \cdot i_d^p(k+1) + \left(1 - \frac{T_s R}{L} - \frac{\lambda T}{AL}\right) i_d(k) \\ &\quad + \left(1 - \frac{\lambda T_s}{AL}\right) \cdot \omega T_s i_q(k) \end{aligned} \quad (31)$$

where $A = [i_d^p(k) - i_d(k)] \cdot u_d(k-1)$ and it denotes known current and voltage information, the value of which is constant at each sampling moment.

Due to the relatively small magnitude of T/L [25], (31) can be simplified as

$$\begin{aligned} i_d^c(k+1) &= \frac{\lambda T_s}{AL} \cdot i_d^p(k+1) + \left(1 - \frac{\lambda T_s}{AL}\right) i_d(k) \\ &\quad + \left(1 - \frac{\lambda T_s}{AL}\right) \cdot \omega T_s i_q(k). \end{aligned} \quad (32)$$

Since the part of (32) containing ω changes very slowly, this speed can be considered as a disturbance. The transfer function (33) can be obtained after z transformation of (32)

$$\frac{i_d^c(z)}{i_d^p(z)} = \frac{\frac{\lambda T_s}{AL} \cdot z}{z - (1 - \frac{\lambda T_s}{AL})}. \quad (33)$$

Based upon (33), the characteristic root of the system can be expressed as $1 - \lambda T / AL$. Similar to [31], the root locus method of discrete system is adopted in this article. The necessary and sufficient condition for the stability of the discrete control

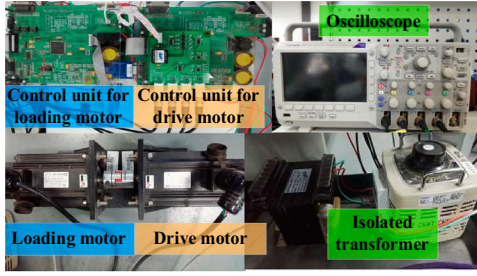


Fig. 6. Experimental platform.

 TABLE I
UNITS FOR MAGNETIC PROPERTIES

Name	Symbol	Value
DC Bus Voltage	U_{dc} (V)	310
Resistance	R (Ω)	3.18
Inductance	L (mH)	8.5
Flux-linkage	ψ_f (Wb)	0.325
Rated Power	P_N (kW)	1.25
Rated Torque	T_N (N·m)	5
Rated Speed	n_N (r/min)	2000
Rated Current	I_N (A)	5
Rotational inertia	J (kg·m ²)	0.00046
Polar Logarithm	p	2

system (DCS) is that the poles of the system are in the unit circle of the z -plane, which can be expressed as

$$-1 < \frac{AL - \lambda T_s}{AL} < 1. \quad (34)$$

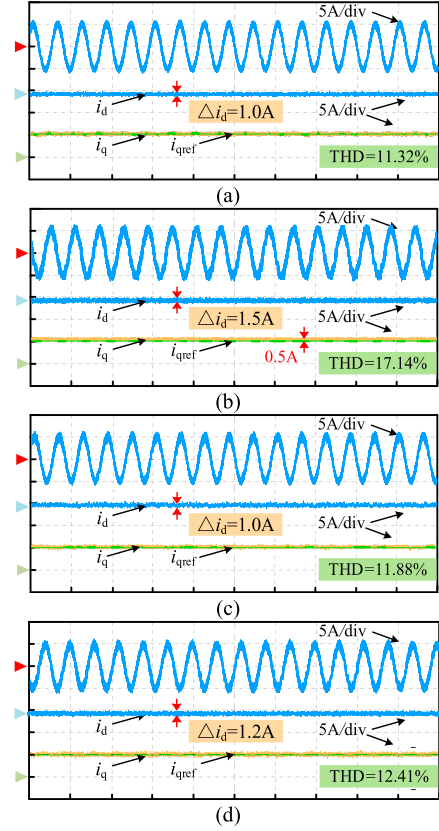
Therefore, the weight coefficient (λ) range to keep the system stable is

$$0 < \lambda < \frac{2AL}{T_s}. \quad (35)$$

In this article, the accurate inductance value (L) is $8.5e^{-3}$, the sampling period (T_s) is $6.666e^{-4}$ and the order of magnitude of the information (A) of current and voltage is $10e^2$. Based on the above data, it is possible to obtain the order of magnitude of the maximum value of weight coefficient (λ) that makes the system stable is $1e^4$. Thus, the weight coefficients (λ) of magnitude $1e^3$ selected in this article are able to maintain the stability of the system.

IV. EXPERIMENTAL RESULTS

To authenticate the feasibility and effectiveness of the proposed MPF-MPCC methodology in this study, a hardware system based on TI's DSP chip TMS320F28335 is built as depicted in Fig. 6. This hardware setup primarily encompasses the drive motor, the load motor and their respective control units. Then, the proposed MPF-MPCC methodology, the conventional MPCC methodology (Methodology-I), the MFPC based on ultra-local model methodology in [28] (Methodology-II) and the methodology in [30] (Methodology-III) are compared in the various operation condition. Table I shows the PMSM parameter and the control frequency adopted by the system is fixed at 15 kHz.


 Fig. 7. Experimental data of 500 r/min with 5 N-m. (a) Methodology-I with accurate parameter. (b) Methodology-I with parameter mismatch ($R_0 = 2R$, $L_0 = 2L$, and $\psi_{f0} = 2\psi_f$). (c) Proposed MPF-MPCC methodology. (d) Methodology-II.

A. Experimental Data in the Steady-State Condition

Within the data obtained from the steady-state experiment, a comparison is made among the total harmonic distortion (THD) of the phase current, third harmonic component (THC), the ripples of the current for d-axis and the static error of the current for q-axis. In the following content, the motor operating condition of 500, 1000, 1500, and 2000 r/min with the rated load torque (5 N-m) is shown and analyzed. The data are also compared with that of the Methodology-I, Methodology-II, and Methodology-III.

Fig. 7 illustrates the experimental data obtained through the three distinct control methodologies under the same operating condition for PMSM. As illustrated in Fig. 7(a), when the Methodology-I is applied to the PMSM operating at 500 r/min with a load torque of 5 N-m, the phase current THD is 11.32%, the current ripple in the d-axis is 1 A, and the current in the q-axis exhibits no static error. However, when the parameters are mismatched, the controllability of Methodology-I suffers from a poor effect, and its phase current THD and d-axis current ripple rise to 17.14% and 1.5A, respectively, and the q-axis current generates a static error, depicted as Fig. 7(b). This suggests that Methodology-I is not characterized by parameter robustness. The proposed MPF-MPCC indicates that the phase current THD is 11.88% and that the current ripple in d-axis is 1A, with no static error in q-axis current [Fig. 7(c)]. The data from

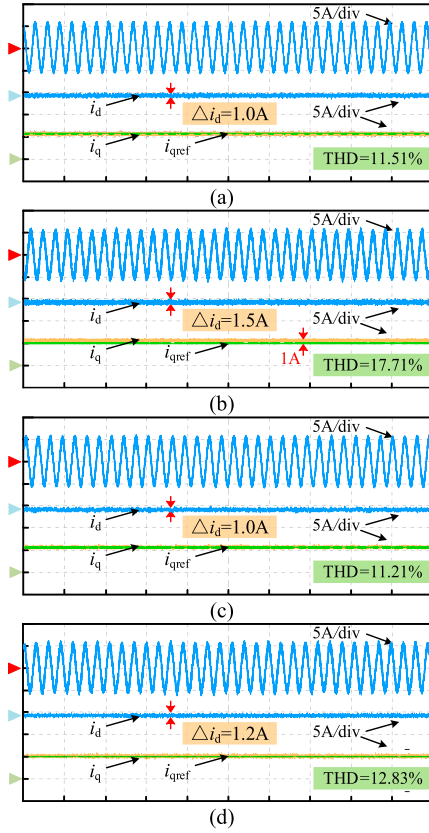


Fig. 8. Experimental data of 1000 r/min with 5 N-m. (a) Methodology-I with accurate parameter. (b) Methodology-I with parameter mismatch ($R_0 = 2R$, $L_0 = 2L$, and $\psi_{10} = 2\psi_f$). (c) Proposed MPF-MPCC methodology. (d) Methodology-II.

Fig. 7(a)–7(c) demonstrates that the proposed MPF-MPCC methodology, in the absence of motor parameters, can achieve a comparable controllability to the Methodology-I with accurate parameter and exhibit parameter robustness that the Methodology-I lacks. Additionally, the phase current THD and d-axis current ripple of Methodology-II are 12.41% and 1.2A, respectively. The proposed MPF-MPCC methodology exhibits superior control effect in comparison to Methodology-II.

Figs. 8–10 respectively illustrate the experimental data obtained from three distinct control methodologies for PMSM under the various speeds, including of 1000, 1500, and 2000 r/min, and with the same load torque of 5 N-m. The observation from Figs. 8(a)–10(a) is that the phase current THD of Methodology-I with accurate parameter is, 11.51%, 11.57%, and 11.82%, respectively. However, when parameters are mismatched, the controllability of the Methodology-I is not effective, resulting in a phase current THD of 17.71%, 17.09%, and 16.99%, respectively. Moreover, the d-axis current ripple increases from 1A to 1.5A for accurate parameters, and the q-axis current also produces static error [Figs. 8(b)–10(b)]. It is noteworthy that the proposed MPF-MPCC methodology indicates that the phase current THD is 11.21%, 11.56%, and 11.52%, respectively. Additionally, the current ripple in d-axis is 1A, with no static error in q-axis current [Figs. 8(c)–10(c)]. The experimental data indicates that the proposed MPF-MPCC methodology at full speeds, in the absence of motor parameters, can achieve a comparable

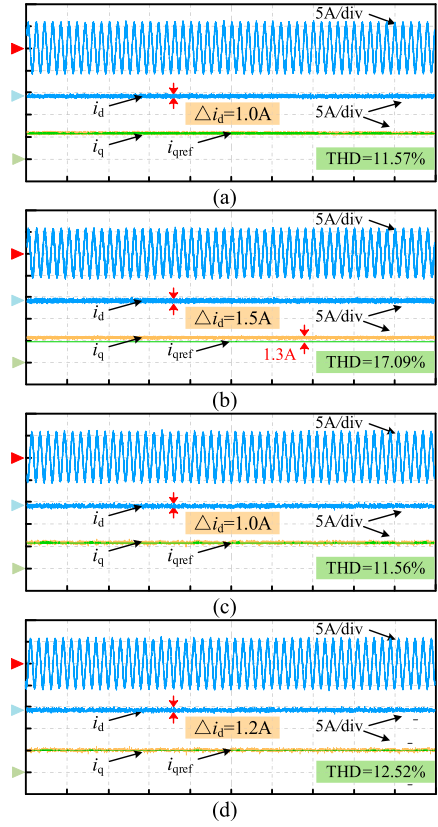


Fig. 9. Experimental data of 1500 r/min with 5 N-m. (a) Methodology-I with accurate parameter. (b) Methodology-I with parameter mismatch ($R_0 = 2R$, $L_0 = 2L$, and $\psi_{10} = 2\psi_f$). (c) Proposed MPF-MPCC methodology. (d) Methodology-II.

controllability to the Methodology-I with accurate parameter and exhibit parameter robustness that the Methodology-I lacks. Furthermore, the control effect is comparable to that of methodology-I with accurate parameter and markedly superior to that of methodology-I with parameter mismatch. The observation from Fig. 8(d)–10(d) is that the phase current THD of Methodology-II, 12.83%, 12.52% and 12.62%, respectively. And, the d-axis current ripple is 1.2A at various speeds. Likewise, the proposed MPF-MPCC methodology exhibits superior control effect in comparison to Methodology-II at full speeds.

To further demonstrate the superiority of the proposed MPF-MPCC methodology in comparison to Methodology-I, the phase currents of the two methodologies are analyzed by fast Fourier transform (FFT) and the respective third harmonic components are obtained. This is shown in Fig. 11. The THCs of Methodology-I with accurate parameters were found to be 0.48%, 0.42%, 0.61%, and 0.23%, respectively, through the FFT analysis of phase currents. In contrast, the THCs of Methodology-I with parameter mismatch were found to be 0.7%, 0.34%, 0.53%, and 1.58%, respectively. This evidence demonstrates that parameter mismatch not only affects the control performance of the drive system but also leads to an increase in the THC, which can result in elevated motor temperature, and additional energy consumption of the motor. Moreover, the THCs of the proposed MPF-MPCC methodology are, respectively, 0.38%, 0.29%, 0.23%, and 0.16%. The

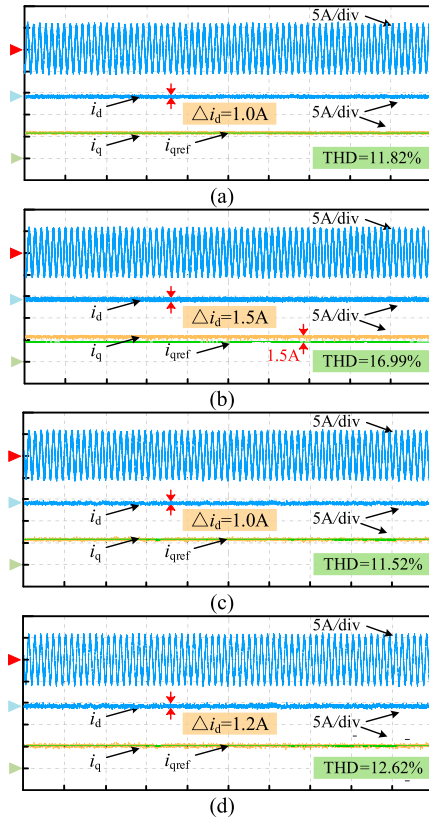


Fig. 10. Experimental data of 2000 r/min with 5 N-m. (a) Methodology-I with accurate parameter. (b) Methodology-I with parameter mismatch ($R_0 = 2R$, $L_0 = 2L$, and $\psi_{f0} = 2\psi_f$). (c) Proposed MPF-MPCC methodology. (d) Methodology-II.

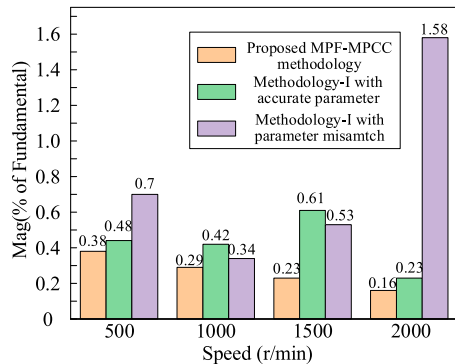


Fig. 11. Bar graph of third harmonic components for different methods at different speed conditions.

results demonstrate that the proposed MPF-MPCC methodology not only maintains satisfactory control performance in comparison to Methodology-I, but also exhibits a reduction in THD.

Furthermore, the observations from Figs. 7(c)–10(c) highlight that as the rotational speed escalates, there is a nonexistent static error of q-axis current in the proposed MPF-MPCC methodology, whereas the static error of q-axis current in the Methodology-I with parameter mismatch [Figs. 7(b)–10(b)] demonstrates a gradual increase. This distinction is further highlighted in Fig. 12, which presents a comparison of the static

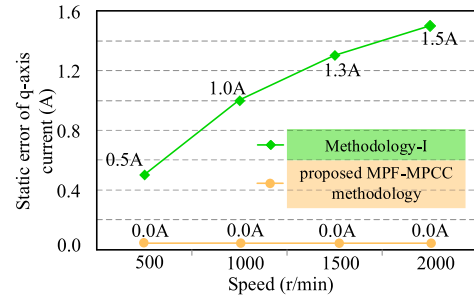


Fig. 12. Line chart of q-axis current static error if the proposed MPF-MPCC methodology and the methodology-I with load torque (5 N-m) at different speeds.

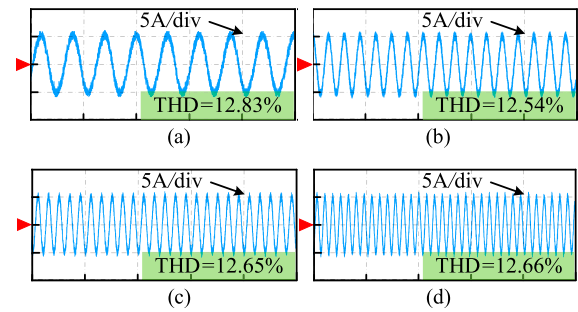


Fig. 13. Phase current THD of the methodology-III at various speeds with 5 N-m. (a) 500 rpm. (b) 1000 rpm. (c) 1500 rpm. (d) 2000 rpm.

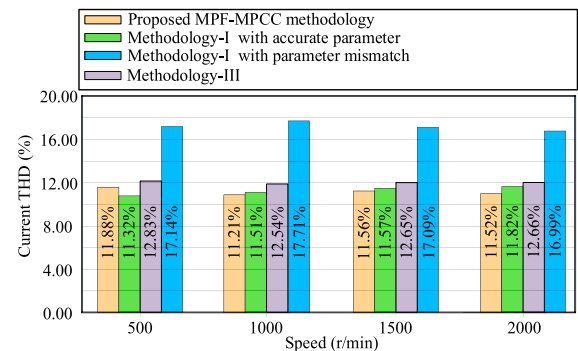


Fig. 14. Bar chart of phase current THD of proposed MPF-MPCC methodology, methodology-I, and methodology-III.

error for the q-axis current across various speeds. Consequently, the proposed MPF-MPCC methodology effectively prevents the occurrence of q-axis current static error resulting from parameter mismatch.

In addition, for a more pronounced visualization of the steady-state controllability for the proposed MPF-MPCC methodology at full speed with the rated load of the motor, a comparison results of the phase current THD across various speeds is presented in Fig. 14. Moreover, the waveform and THD of the methodology-III are depicted in Fig. 13. A discernible observation from Fig. 14 is that the THD of the proposed MPF-MPCC methodology is evident lower as compared to Methodology-I with parameter mismatch as well as Methodology-III and is almost indistinguishable from that of Methodology-I with accurate parameter. This further illustrates that the MPF-MPCC

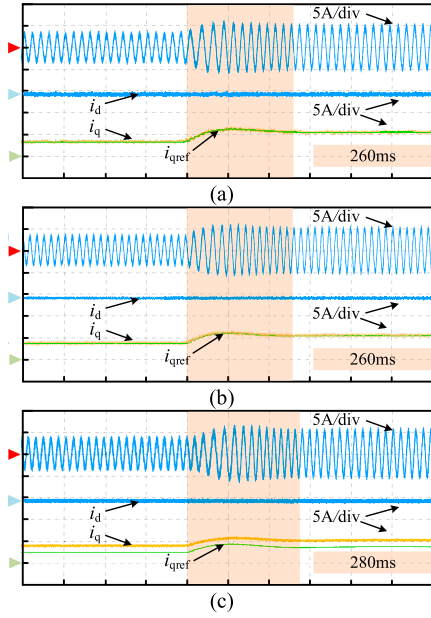


Fig. 15. Experimental data of 1500 r/min with load torque from 3 to 5 N-m. (a) Proposed MPF-MPCC methodology. (b) Methodology-I. (c) Methodology-I with parameter mismatch.

methodology, due to the absence of any motor parameters in the model, exhibits not only robust performance but also excellent steady-state control capabilities.

The above analysis can confirm that the proposed MPF-MPCC methodology exhibits greater robustness than Methodology-I at full speeds. Furthermore, it is capable of achieving the control effect of Methodology-I with parameter mismatch while exhibiting a lower THC. Besides, compared to methodology-II and methodology-III, the proposed MPF-MPCC methodology has lower THD in the same condition.

B. Experimental Data in the Dynamic-State Condition

Fig. 15 shows the dynamic experimental data of the three different methodologies when PMSM runs at 1500 r/min with the load torque changes from 3 to 5 N-m. The data depicted in Fig. 15 reveals that the dynamic response times for PMSM applied by the distinct methodologies are nearly identical in scenarios involving abrupt changes in torque loads. Nevertheless, a significant observation is that the PMSM remains the static error of q-axis current when the load torque experiences an abrupt shift from 3 N-m to 5 N-m within Methodology-I with parameter mismatch.

In addition, Fig. 16 illustrates the dynamic experimental data of the proposed MPF-MPCC methodology and Methodology-I when PMSM operates at speeds ranging from 1000 to 1500 r/min, with a 5 N-m load torque. The insights gleaned from Fig. 16 indicate that the two methodologies exhibit comparable dynamic response times when confronted with abrupt changes in speed. It is noteworthy that the q-axis current error persists in Methodology-I with parameter mismatch when the speed abruptly changes from 1000 to 1500 r/min, resulting in a longer dynamic response time due to parameter mismatch. Furthermore, the proposed MPF-MPCC methodology exhibits a dynamic response time of 250 ms,

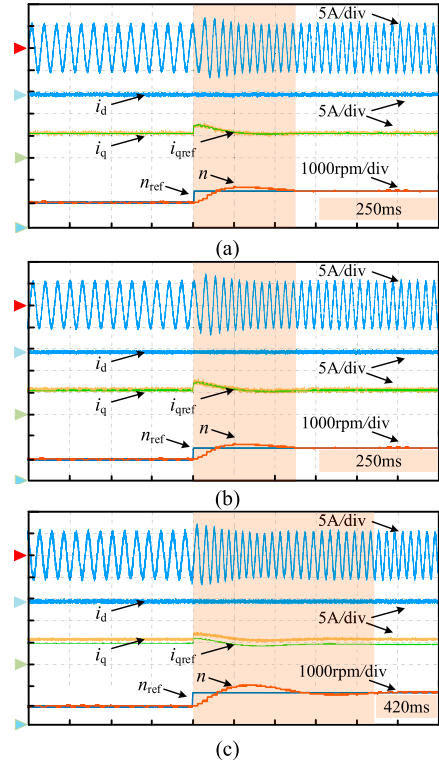


Fig. 16. Experimental data of speed from 1000 to 1500 r/min with 5 N-m load torque. (a) Proposed MPF-MPCC methodology. (b) Methodology-I with accurate parameter. (c) Methodology-I with parameter mismatch.

which aligns with the Methodology-I with accurate parameter. This demonstrates that the proposed MPF-MPCC methodology is capable of circumventing the adverse effects associated with Methodology-I, which are a consequence of parameter mismatch.

In conclusion, the above experimental data affirm that the proposed MPF-MPCC methodology has almost the same dynamic-state controllability as the Methodology-I and can avoid the q-axis current error and obtain faster response time for speed variation to achieve better dynamic-state performance when the parameters are mismatched.

C. Comparison of the Characteristics of the Different Methodologies

To demonstrate the superiority of the proposed methodology, the characteristics of proposed methodology, methodology-I, methodology-II, and methodology-III are compared in this part, as depicted in Table II. In comparison to the methodology-I, the proposed methodology entails a $3.48 \mu\text{s}$ increase in computational burden. However, this is offset by enhanced parameter robustness, which is a highly valuable outcome. Furthermore, the computational burden of the proposed methodology is reduced by $2.07 \mu\text{s}$ in comparison to methodology-III. Compared to methodology-II, the proposed methodology as fewer system parameter (only one) that need to be tuned and has no additional observer design required. It is concluded that the proposed methodology requires only a low extra computational burden to achieve the system robustness improvement and is simple to implement.

TABLE II
PROGRAM EXECUTION TIME

	Methodology-I	Methodology-II	Methodology-III	Proposed Methodology
Clock cycles	2941	3212	3772	3463
Algorithm execution time	19.6 μ s	21.41 μ s	25.15 μ s	23.08 μ s
Whether motor parameters are required	Yes (R , L , ψ_f and so on)	No	No	No
Parameters to be designed or tuned	\	2	2	1
Whether an observer is required	\	Yes	No	No

V. CONCLUSION

To remove the effect of parameter mismatch within the prediction model on the PMSM drive system, an MPF-MPCC methodology is proposed in this study. And, the primary contributions are outlined as follows:

- 1) An estimation factor (M) is designed to substitute the inductance term (T_s/L) in the model. Besides, by eliminating resistance and flux-linkage, an updated current prediction model devoid of the existence in any motor parameters is established.
- 2) Through the designed simple control rhythm, the estimation factor is calculated in real time using the predictive errors of d-axis current and applied to the updated current prediction model enabled predictive accurately current in the absence of any motor parameters. And the theoretical range of weight coefficients in the control law is given.
- 3) The preceding experimental results permit the following summary down. First, the proposed MPF-MPCC methodology is relatively straightforward to implement and does not necessitate the design of an additional observer and the tuning of a greater number of system parameters than Methodology-II. In comparison to Methodology-I, the proposed MPF-MPCC methodology exhibits comparable steady-state and dynamic control performance, with the accuracy of the parameters. However, when the parameters are mismatched, the dynamic steady-state and dynamic-state performance of the proposed MPF-MPCC methodology is superior to that of Methodology-I. Moreover, in comparison to Methodology-III, the proposed MPF-MPCC methodology exhibits a reduced time burden and superior steady-state control performance. This demonstrates that proposed MPF-MPCC methodology does not necessitate the input of any motor parameters, thereby effectively and simply enhancing the parameter robustness.

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